

Material Safety Data Sheet

SECTION 1 IDENTIFICATION

Product Name: Spirodiclofen 240g/L SC

Chemical name: 3-(2, 4-dichlorophenyl)-2-oxo-1-oxaspiro[4.5]dec-3-en-4-yl 2,2-dimethylbutanoate

Other Means of identification: /

Recommended use of the chemical and restrictions on use: This material is a kind of pesticide.

SECTION 2 HAZARDS IDENTIFICATION

Classification of the substance or mixture:

Sensitisation, skin Category 1

Hazardous to the aquatic environment, long-term hazard Category 4

GHS Label elements, including precautionary statements:



Signal word: Warning

Hazard statement(s): May cause an allergic skin reaction. May cause long lasting harmful effects to aquatic life.

Precautionary statement(s):

Prevention:

Avoid breathing dust / fume / gas / mist / vapors / spray. Contaminated work clothing should not be allowed out of the workplace. Wear protective gloves. Avoid release to the environment.

Response:

If on skin: Wash with plenty of water. If skin irritation or rash occurs: Get medical advice/attention.

Specific treatment (see safety data sheet). Take off contaminated clothing and wash it before reuse.

Storage: /

Disposal: Dispose of contents/container in accordance with local / regional / national / international regulations.

Other hazards which do not result in classification: /

SECTION 3 COMPOSITION/INFORMATION ON INGREDIENTS

Component	CAS No.	Concentration, w/v, g/L
Spirodiclofen	148477-71-8	240
Adjuvants	/	760

SECTION 4 FIRST AID MEASURES

Description of necessary first aid measures

If inhaled: If breathed in, move person into fresh air. If not breathing, give artificial respiration. Consult a physician.

In case of skin contact: Wash off with soap and plenty of water. Consult a physician.

In case of eye contact: Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.

If Ingestion: Rinse mouth with water. Consult a physician.

Most important symptoms/effects, acute and delayed: /

Indication of immediate medical attention and special treatment needed, if necessary: /

SECTION 5 FIREFIGHTING MEASURES

Suitable extinguishing media: Use foam, chemical powder, carbon dioxide or water spray.

Special hazards arising from the chemical: This material may decompose and burn at high temperature and fire and release toxic fumes.

Special protective actions for fire-fighters: Wear self-contained breathing apparatus for firefighting if necessary. Use water spray to cool unopened containers. Move the container from the fire to the open area as much as possible. Then the appropriate extinguishing agent should be selected according to the fire reason.

SECTION 6 ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures: Use personal protective equipment. Avoid breathing vapors, mist or gas. Ensure adequate ventilation. Remove all sources of ignition. Evacuate personnel to safe areas.

Environmental precautions: Prevent further leakage or spillage if safe to do so. Do not let product enter drains.

Methods and materials for containment and cleaning up: Contain spillage and then collect with pump and place in a clean container for disposal according to local regulations.

SECTION 7 HANDLING AND STORAGE

Precautions for safe handling: Wear protective gloves / eye protection / face protection / protective clothing. Avoid contact with skin and eyes. Avoid inhalation of vapour or mist. Keep away from sources of ignition - No smoking.

Conditions for safe storage, including any incompatibilities: Store in cool place. Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage. Keep away from flammable materials and oxidizers.

The storage area shall be equipped with suitable materials for shelter and leakage.

SECTION 8 EXPOSURE CONTROLS/PERSONAL PROTECTION

Control parameters: /

Appropriate engineering controls: Local exhaust ventilation or a process enclosure ventilation system may be required.

Individual protection measures

Eye/face protection: Safety glasses with side shields. Chemical goggles. Contact lenses may pose a special hazard; soft contact lenses may absorb and concentrate irritants.

Skin protection: Wear chemical protective gloves, e.g. PVC. Wear safety footwear or safety gumboots, e.g. Rubber. Impervious clothing,

Respiratory protection: Selection of the class and type of respirator will depend upon the level of breathing zone contaminant and the chemical nature of the contaminant.

Thermal hazards: /

SECTION 9 PHYSICAL AND CHEMICAL PROPERTIES

Appearance (physical state, colour etc)	White viscous liquid
pH	5.0-8.0
Suspensibility	90% Min.
Melting point/freezing point	N/A
Initial boiling point and boiling range	N/A
Flash point	Non-flammable in nature
Evaporation rate	N/A
Flammability (solid, gas)	N/A
Upper/lower flammability or explosive limits	N/A
Vapour pressure	N/A
Vapour density	N/A
Solubility(ies)	Soluble in water.
Partition coefficient: n-octanol/water	N/A
Auto-ignition temperature	N/A
Decomposition temperature	N/A

SECTION 10 STABILITY AND REACTIVITY

Reactivity: /**Chemical stability:** This material is stable in normal temperature.**Possibility of hazardous reactions:** /**Conditions to avoid:** Spark, static electricity and high temperature.**Incompatible materials:** Flammable materials and oxidizers.**Hazardous decomposition products:** Oxycarbides, chlorides, etc.

SECTION 11 TOXICOLOGICAL INFORMATION

Information on the likely routes of exposure: Inhaled, swallowed, skin, eyes.**Symptoms related to the physical, chemical and toxicological characteristics:** /**Acute health effects:** Accidental ingestion of the material may be harmful and cause cough and throat pain. Oral intake may cause headache, giddiness, vomit and other symptoms. This material may produce skin and eyes irritation.**Chronic health effects:** Repeated or prolonged exposure to this material may cause skin allergy.**Numerical measures of toxicity(such as acute toxicity estimates):**Acute Oral Toxicity (Rat): $LD_{50} = 5000$ mg/kg body weight, Classified in GHS Category 5 or unclassified.Acute Dermal Toxicity (Rat): $2000 < LD_{50} \leq 5000$ mg/kg body weight, Classified in GHS Category 5.Acute Inhalation Toxicity (Rat): $LC_{50}(4h) > 4.12$ mg/L, Could not be classified according to GHS.

Acute Eyes Irritation (Rabbit) : Non-irritant, Could not be classified according to GHS

Acute Dermal Irritation (Rabbit) : Non-irritant, Could not be classified according to GHS

Skin Sensitization(Guinea Pig) : Non-sensitiser

ADI: 0.015 mg/kg body weight.

SECTION 12 ECOLOGICAL INFORMATION

Toxicity:

Birds (bobwhite quail) :	Acute oral $LD_{50} > 2000$ mg/kg body weight. Dietary $LC_{50}(5d) > 5000$ mg/kg feed.
Fish (rainbow trout) :	$LC_{50}(96 h) > 0.035$ mg/L.
Bees:	Oral $LD_{50} > 196 \mu\text{g}/\text{bee}$; Contact $LD_{50} > 200 \mu\text{g}/\text{bee}$.
Daphnia:	$EC_{50}(48 h) > 0.051$ mg/L.
Algae (Pseudokirchneriella subcapitata) :	$ErC_{50}(72 h) > 0.06$ mg/L; $E_bC_{50}(72 h) > 0.06$ mg/L.
Worms(Eisenia foetida):	$LC_{50} > 1000$ mg/kg soil.

Bioaccumulative potential: /**Mobility in soil:** /**Other adverse effects:** /

SECTION 13 DISPOSAL CONSIDERATIONS

Disposal methods: Burial in a land-fill specifically licensed to accept chemical. Reuse of broken container is forbidden.

SECTION 14 TRANSPORT INFORMATION

UN Number: 3082.

UN proper shipping name: ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.

Transport hazard class(es) : 9.

Packing group, if applicable: III.

Environmental hazards: /

Special precautions for user: /

SECTION 15 REGULATORY INFORMATION

Regulations: This Material Safety Data Sheet is in compliance with the following national standards: GB 16483-2008, GB 13690-2009, GB/T 15098-2008, GB 18218-2009, GB 15258-2009, GB 6944-2012, GB 190-2009, GB 191-2009, GB 12268-2008, GA 57-1993, GBZ 2-2007 as well as the following national regulations: Dangerous Goods Transport Administrative Regulation [Published by the Ministry of Railways, 2008], Dangerous Chemicals Safety Administrative Regulation [Published by the State Council, 2011].

SECTION 16 OTHER INFORMATION

All information and instructions provided in this Material Safety Data Sheet (MSDS) are based on the current state of scientific and technical knowledge at the date indicated on the present MSDS and are presented in good faith and believed to be correct. This information applies to the product as such. In case of new formulations or mixes, it is necessary to ascertain that a new danger will not appear. It is the responsibility of persons on receipt of this MSDS to ensure that the information contained herein is properly read and understood by all people who may use, handle, dispose or in any way come in contact with the product. If the recipient subsequently produce formulations containing this product, it is the recipients sole responsibility to ensure the transfer of all relevant information from this MSDS to their own MSDS.